Non-canonical agreement in early grammars[1]

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Abstract

Children acquiring a diverse range of unrelated languages commonly go through a stage, typically between the ages of 2 and 3, where they use bare, infinitival or default forms instead of the agreeing forms required in adult grammar. Whether such under-agreement errors reflect non-adult syntax for agreement or incomplete morphological knowledge has been a topic of debate for decades (Brown 1973, Radford 1990, Rizzi 1994, Wexler, 1998, a.o.). These errors are interesting developmentally, since children otherwise make few grammatical errors, and theoretically, given the central role of agreement in understanding the combinatorial mechanisms of natural language.

In this paper, we approach this classic developmental problem from a new angle, by examining the acquisition of 'non-canonical' agreement constructions, where a non-standard element occupies the subject position, and agreement is with a post-verbal nominal. Such constructions are generally thought to be syntactically more complex than canonical sentences, in that they involve at least two dependencies — one that is responsible for subject movement, and a second that is responsible for the verbal morphology. In these environments, we find a striking reemergence of agreement problems, now in a population who otherwise command the relevant morphology and use it in adult-like ways in canonical contexts. Two elicited repetition studies found that children (ages 3;0–5;11) routinely produced singular agreement with plural associates in non-canonical constructions. We take this over-production of singular to reflect a 'late' under-agreement stage, where children default to singular morphology as a result of failing to agree. Indirectly, these results support syntactic explanations of toddlers' under-agreement and illustrate how the protracted acquisition of agreement tracks syntactic complexity.

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The study of early language acquisition promises to shed light on the nature of human languages by revealing the hypothesis space that learners navigate. The near-perfect accuracy of young children with respect to certain linguistic properties — such as the structure-dependence of grammatical rules (e.g., Crain & Nakayama (1987), Shi et al. (2020)) — has been key evidence for the existence of innate biases that shape the acquisition process. Equally revealing, however, are cases where children's linguistic knowledge is conspicuously lacking with relation to some phenomena for which they have abundant input. Such gaps in knowledge may reflect aspects of grammar where learning is guided by internal principles, rather than being fully determined by the input.

A particularly well-documented example of such a gap is the phenomenon of "under-agreement" in early child speech. Children acquiring a wide variety of unrelated languages go through a stage, typically between the ages 2 and 3, where they omit verbal agreement that is obligatory in adult grammar. Prominent linguistic analyses of the errors attribute them to the late maturation of components involved in building agreement dependencies (e.g., Wexler (1998)). If on the right track, this account has bearing not only on our understanding of language acquisition, but also on central questions in syntactic theory. Contemporary syntactic theory aims to reduce all non-local dependencies, from ϕ-agreement to *wh*-dependencies, to the abstract operation Agree (Chomsky 2000, Chomsky 2001 et seq.; see also Deal (2023)). If certain Agree relations indeed show a distinct developmental pattern from others, this would provide important data for theorizing about which aspects of the combinatorial apparatus can and cannot be unified.

At the same time, the claim that under-agreement reflects a syntactic deficit is not easily settled, as competing non-syntactic alternative explanations of the phenomenon remain viable. The language generated by the relevant special population —young children at the earliest stages of language production— is inherently sparse and messy, making it hard to test fine grained theoretical predictions. In this article, we present novel evidence that bolsters the syntactic position. We show that children continue to have difficulties with agreement far longer than previously thought, late into their preschool years. We refer to this previously undocumented phenomenon as 'late under-agreement', distinguishing it from the 'classic under-agreement' stage in toddlerhood. Strikingly, these children no longer exhibit the classic under-agreement errors, but show agreement failures in specific environments where the target is post-verbal. What makes these configurations critical — and what explains children's errors — is that these post-verbal agreement configurations involve an increase in the number of Agree dependencies.

We take these findings to demonstrate that facility with a central aspect of structure-building develops piece-by-piece, with each bit of added complexity leading to increased difficulty and more errors. In turn, the results also demonstrate how individual operations that are posited in the grammar — e.g. Agree relations — reveal their psychological reality over the course of development.

1 Under-agreement in toddler grammars

Utterances like (1) are characteristic of child English, with toddlers using the bare verb form in place of the adult-like inflected variants. In other languages, children tend to produce the infinitival or default third-singular morphology in place of the expected agreeing forms (2)-(4).

(1) English
  a. Cromer wear glasses (Eve 2;0)
  b. Mumma ride horsie (Sarah 2;6)

(2) Dutch
  Earst kleine boekje lezen *(Hein, 2;6)*
  first little book read-ING
  'First (I/we) read little book.'

(3) French
  a. *Dormir petit bébé (Daniel, 1;11)*
  sleep-INF little baby
  'Little baby sleeps.'
  b. *J'est pas mechant (Tom, 2;4)*
  I.be.3SG not mean
  'I is not mean.'

(4) Spanish
  a. *Yo quiere hacerlo* *(Eduardo, 3;0)*
  I want.3SG do.INF.CL3SG
  'I wants to do it'

Adding to the puzzle, under-agreeing forms often coexist with adult-like forms, indicating that toddlers have not simply failed to learn the relevant inflectional morphology. Finally, these errors follow a fairly consistent time-course: they emerge around age 2, when children begin producing multi-word utterances, and largely disappear by age 3, at which point adult-like agreement becomes the norm.

Under-agreement errors have been the focus of inter-disciplinary research for decades (Brown 1973, Radford 1990, Hoekstra & Hyams 1998, Snyder & Bar-Shalom 1998, Wexler 1998, Bar-Shalom & Snyder 1999, Rizzi 1993, Phillips 2010, a.o.). An intriguing proposal that has emerged is that toddlers' errors reflect late maturation of syntactic components or processes crucial in agreement (Poeppel & Wexler 1993, Rizzi 1993, Schütze & Wexler 1996, Wexler 1998). On this view, toddlers' erroneous sentences have a non-adult underlying structure, which is missing projections, heads, or features that are critical for triggering ϕ-agreement. An adult-like morphology then outputs this structure. For example, Schütze & Wexler (1996) and Wexler (1998) argue that under-agreeing forms are allowed in children's grammars because agreement-relevant features on T are optionally omitted. When these features are missing, syntactic operations triggered by those features (e.g. search for a matching goal) and the consequent structural changes (e.g. feature-valuation) never happen. Absent a specified set of feature-values, there is nothing for the morphology to spell out, and a default form — null, infinitival or 3Sg — is inserted.

On this proposal, a central aspect of grammar, the mechanism responsible for ϕ-agreement, is not yet fully in place in toddlerhood. Support for this position comes from the observation that child utterances with under-agreement often share syntactic properties with adult clauses lacking agreement (e.g., embedded infinitives) (Pierce 1992, Poeppel & Wexler 1993). For example, in child French, non-agreeing verbs appear to the right of negation, mirroring non-agreeing verbs in adult French embedded clauses. Similarly, in child German, non-agreeing verbs are clause-final, rather than in second position, as in adult German infinitives.

But there are non-syntactic explanations for these phenomena as well. For instance, emergentist approaches claim that productive morphology is the result of an extended, gradual learning process (e.g., Kaupiche & Rinke, 2008). Initially, children's knowledge of finite morphology is lacking, and their early inflected productions are therefore unanalyzed chunks. To learn which parts of words are morphemic, children have to compare similar, partly overlapping forms and correlate morphological overlap with differences in meaning and use. These accounts, however, are difficult to reconcile with growing evidence that even during periods where children's own productions are non-adult, they show sensitivity to correct and incorrect agreement (see Shi et al. (2020) for French; Lukyanenko & Fisher (2016) for English).

On another non-syntactic alternative, the problem is not about morphological *learning* but morphological *performance*. Phillips (2010) argues that both syntactic and morphological knowledge are in place by toddlerhood, so children's finite utterances reflect genuine competence. What they struggle with is deploying this knowledge online. These implementation issues may be more dire for children acquiring languages like English and Swedish, with many-to-one mappings of morphological forms to syntactic features. Learners of these languages may need time before accessing and retrieving these forms become fully automatic.

Phillips draws on a well-established generalization that children learning languages with richer inflectional paradigms tend to make fewer errors with inflection (Slobin 1982, Hyams 1986). For instance, children acquiring languages with rich verbal paradigms (like Spanish, Italian, and Hebrew) produce more consistent verbal inflection than those learning German, French, or Dutch, which have fewer person/number distinctions. Children acquiring languages with minimal verbal morphology, such as English and Swedish, show the highest error rates. This cross-linguistic variation is explained if toddlers learning richly inflected languages gain more practice, leading to stronger entrenchment and easier retrieval of morphological forms.

Both types of accounts are *maturational* hypotheses, in that they assume that some cognitive capacity necessary for adult-like behavior is unavailable until a later developmental stage. But, they differ on what exactly matures. The syntactic account postulates maturation of the *linguistic* capacity, whether in *knowledge* of some relevant constraint Rizzi (1993), or in the *ability to execute* some linguistic capacity Wexler (2011). The morphological performance account attributes the error to late maturation of mental capacities outside of the language faculty, e.g. limitations of memory capacity or overall processing speed. But determining the true source of these errors is challenging. The data are inherently limited, and the population in question has more general issues with limited working memory and lexical retrieval (Gershkoff-Stowe & Smith 1997, Dapretto & Bjork 2000), making it hard to identify the source of *any* error.

Here, we approach this question by looking at older children (3-to-5-year-olds), who are no longer in the classic under-agreement stage. These children are more proficient in producing language, and have fewer generalized performance issues. Critically, by 3 years of age, they have begun to reliably use agreeing verb forms in both production and comprehension (Brown 1973, Rice & Wexler 2001, Theakston & Rowland 2009, Lukyanenko & Fisher 2016; Rissman et al. 2013). If, with an increase in syntactic complexity, agreement errors resurface in this population, that would suggest that some aspect of syntax is at the core of under-agreement errors — whether in the maturation of core syntactic competencies, or some abilities that are selectively taxed by syntactic structure building.[2]

Our crucial test environment are constructions involving 'non-canonical' agreement, whose properties allow us to increase the syntactic complexity of sentences without changing the associated morphological demands. In canonical agreement constructions, exemplified in (5), the verb agrees with the subject in ϕ-features. In 'non-canonical' agreement constructions, such as existential-*there* constructions (6) and locative-inversion constructions (7), a non-standard element (expletive *there*, locative PP) occupies the subject position, but the verbal morphology co-varies with a post-verbal nominal.

(5) **Predicative constructions**
Some toys are in the box.

(6) **Existential-*there* constructions**
There are some toys in the box.

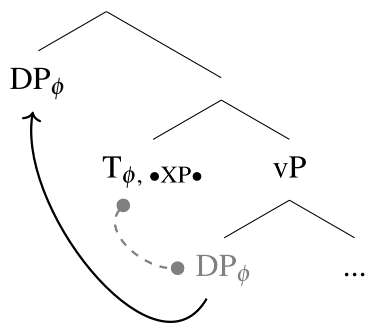
(7) **Locative inversion constructions**
In the box are some toys.

On the surface, all these sentence types appear quite simple; simple enough that a 3-year-old can produce them without issues. Nevertheless the non-canonical constructions are syntactically more involved when it comes to agreement. In canonical agreement scenarios, both T's ϕ-featural needs and its need for a specifier (its EPP property) are satisfied by a single goal: the subject DP, (8). Put differently, both the verbal morphology and subject movement can be thought of as structural consequences of a single Agree relation. In non-canonical agreement constructions, the needs of the probe are distributed across two

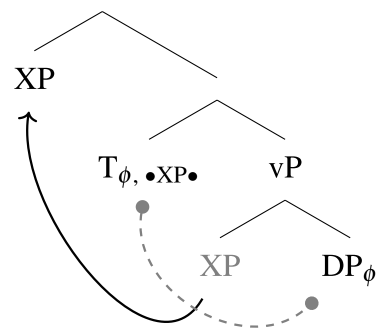
[2]On such hybrid accounts errors would emerge in more complex structures because some processing limitations could interact specifically with syntactic complexity. We discuss this view in more detail in the General Discussion.

goals. What moves to the subject position is *there* in (6) or a locative PP in (7), but what provides ϕ-featural values — and thus controls agreement morphology on the verb — is a lower down DP. On the assumption that Agree is a pre-requisite for movement, these constructions then involve at least two Agree relations; (9). Critically, there is no morphological difference: both canonical and non-canonical constructions draw from the same morphological paradigms.

(8)



(9)



Our main empirical question is about the relationship between syntactic complexity and errors with agreement. If only morphology is at issue, increased syntactic complexity should be irrelevant: errors should not depend on construction, only on whether the child has mastered the morphology. This prediction is most straightforward for emergentist accounts on which under-agreement would simply reflect incomplete morphological knowledge. On the other hand, if (1) under-agreement has syntactic roots, and (2) these syntactic roots are related to syntactic complexity — as defined by the number of Agree relations — then we can predict that even older children should struggle when syntactic complexity is increased. That is, we should be able to observe a new type of under-agreement phenomena: *late* under-agreement. In what follows, we will try to arbitrate between these two hypotheses by probing whether the prediction of *late* under-agreement is borne out.

Is there any reason to expect that late under-agreement exists? The only promising prior evidence we are aware of is a corpus study from Lukyanenko & Miller (2023), which explored the effects of input variation on children's production of agreement. In two studies examining US English caregiver-child interactions in naturalistic speech, they indeed found that children tended to overproduce singular agreement with plural associates. While they did not code for syntactic complexity — as they were approaching the question from a variationist viewpoint — they strikingly found more errors in non-canonical constructions. They also found that caregivers displayed occasional under-agreement, but children's errors were not fully explained by caregiver patterns — children tended to generate drastically more under-agreement than their caregivers. While promising, it is hard to draw strong conclusions from this study alone. Beyond its non-experimental nature, the two children in the sample had data from a wide age-range that spans both classic and late under-agreement stages. Furthermore, because the paper has a different set of theoretical assumptions, the analysis collapses several distinctions that are important for our work (see §3 for further discussion). To zero in on the relevant phenomenon, we need an experimental paradigm that is purpose-made to the relevant morphologically sophisticated 3-5 year-old age group, with tests that manipulate the critical distinctions — and no others.

To do so, we created two Elicited Repetition experiments aimed at directly probing children's production of agreement in canonical and non-canonical sentences. Our first experiment, detailed in §2, looks at *there*-existential sentences. We find evidence for late under-agreement: children prefer default singular morphology in existentials even when the post-verbal DP is plural, and often even correct adult-like plural agreement in such cases. §3 reports on a corpus based analysis of child-directed speech as a follow-up analysis. Our findings demonstrate that the singular preference is unlikely to be driven by the input children receive. In §4, we extend our investigation to another non-canonical agreement construction — locative inversion. Here again, we find an over-production of singular agreement, indicating that late under-agreement indeed tracks syntactic complexity rather than being specific to particular constructions.

2 Experiment 1: *there*-existentials

The first experiment examines *there*-existentials as in (10). These sentences are 'non-canonical' in two ways. First, a non-standard element, the locative-expletive *there*, occupies the subject position. Second, the verb morphology co-varies with the post-copular nominal, the 'associate'.

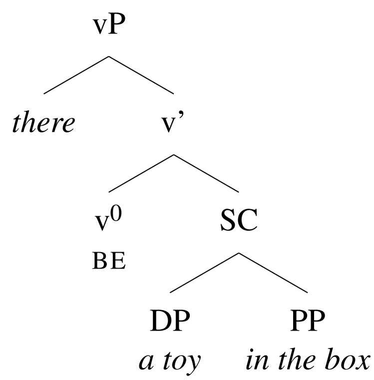
(10) a. There is a toy in the box.
b. There are some toys in the box.

These constructions are ideal starting points because they are frequent in child-directed speech (discussed further in §3) and appear in children's own speech before age 3 (Inoue (1991), Schafer & Roeper (2000)).

Though the literature on the *there*-existential construction within the generative tradition is substantial and there is no real consensus, here we will take the view that existentials are copular sentences with *there* as the subject, the associate nominal as the predicate, and any additional phrase after the associate (e.g. "in the box" in (10); sometimes called the 'coda') as an adjunct or modifier (Williams (1984), McNally (1992), Hazout (2004)). The agreement profile of existentials is a consequence of the locative expletive lacking the full set of ϕ-features necessary to fulfill the needs of T (Chomsky (2000), Hazout (2004), Longenbaugh (2019)).

In existentials and other constructions with *there*, the expletive is merged low, in Spec, vP (Deal, 2009), in a structure as in (11). Finite T in English needs ϕ-features and also a subject. Because *there can* occupy subject position in English, it is a partly suitable goal for T. The expletive cannot, however, satisfy T's ϕ-featural needs, because it lacks (at least) number features, and these still unvalued ϕ-features on T trigger a second search. As a result of this second cycle of Agree, the probe finds the lower DP and copies its values. Thus, we have ϕ-featural covariance between the verb and post-verbal DP.

(11)



We probe children's command of this construction using an Elicited Repetition task, in which they are asked to repeat a structure modeled by the experimenter. This paradigm has been widely used in developmental syntax research (e.g., Fraser et al. (1963), Lust et al. (1987), McDaniel et al. (1998), Ambridge & Pine (2006), Valian et al. (1996)) and relies on a key assumption: that children only repeat constructions they have in their grammar. When confronted with a structure that their grammar cannot generate, they repair them, sometimes even to non-target-like forms. The paradigm allows us to test children's language generation capacity, in a carefully controlled manner, with constructions that may not appear frequently in their naturalistic speech.

2.1 Methods

All methods and analysis plans were preregistered here: https://osf.io/m7uf8/?view_only=202afbeff6b64d8aa72f278ed6b0ec5a. Methodological and analytical choices were as specified there, unless otherwise noted.

2.1.1 Participants

We analyze results from a sample of 60 children (Mean Age = 4;5, Range = 3;0-5;11), whose dominant language (at least 60% of home use) was English. Participants were recruited from preschools and museums in the MASKED area. We did not collect detailed demographic or language background information beyond overall English exposure. An additional 6 children were tested but excluded for technical problems (4) or failure to complete the study (2). In both experiments, parents gave informed consent and children received a small reward for participation.

2.1.2 Design, Materials and Procedure

Experiment 1 used a 2x2x2 design, crossing Construction (canonical predicative vs. non-canonical *there-existentials*), Associate Number (singular vs. plural) and Verb Agreement (is vs. are). Crossing these three factors resulted in 8 within-subject conditions, half of them ungrammatical. Conditions and sample items are provided in Table 1.

Participants were first introduced to a cartoon character, a "shy" giraffe, who served as the protagonist in this experiment as well as Experiment 2. The giraffe had different containers (e.g. box) in its possession. Children were told that the giraffe has things inside the container, but will reveal them only if both the experimenter and child ask him about what he has. The experimenter would first pose a question, and then the child would need to repeat it. Upon repetition, the object in the container is revealed. The experiment began with a warm-up phase, where the experimenter greeted the giraffe and the child is asked to repeat the greeting.

Figure 1 shows an example trial. Each trial involved a Yes/No question formed from either an existential-*there* construction or a predicative copular construction. The sentences uniformly involved an associate DP (e.g. a toy) and a spatial PP (e.g. in the box). Both the associate DP and form of the copula could also vary across trials.

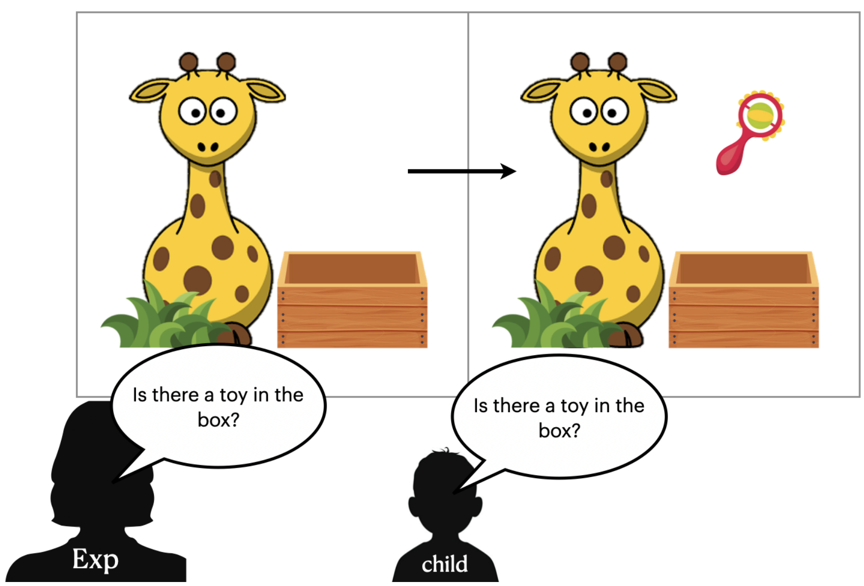


Figure 1: Structure of a trial; Experimenter produces a prompt that the child repeats; upon repetition, concealed object is revealed

There were 2 items per condition, yielding 16 total test trials. We limited each session to 16 test trials to ensure that, especially for our youngest participants, the task remained short enough to maintain attention and reduce fatigue while preserving a fully within-subjects design. Test trials were grouped into blocks by construction, and each participant saw trials in one of four orders with block order counterbalanced. Each testing session lasted between

	Verb Agreement	Associate Number	
		Singular	Plural
Non-canonical	is	Is there a toy in the box?	Is there some toys in the box?
	are	Are there a toy in the box?	Are there some toys in the box?
Canonical	is	Is a toy in the box?	Is some toys in the box?
	are	Are a toy in the box?	Are some toys in the box?

Table 1: Trial types, Experiment 1

20-30 minutes.

For each trial, we measured whether the child's repetition attempt matched the prompt, coded as a binary variable (Repetition Match; 1=Yes, 0=No). Additionally, we coded the type of change they made to the prompt when their repetitions did not match. These changes were categorized into four categories, listed in Table 2.

Type of change	Example
Agreement change	Are there a toy in the box? → Is there a toy in the box?
Associate change	Are there a toy in the box? → Are there some toys in the box?
Construction change	Are there a toy in the box? → Are a toy in the box?
Other	

Table 2: Coding scheme for changes, Experiment 1

If children are adult-like, we expect repetition match to be modulated by grammaticality. That is, they should repeat the experimenter's prompt if it is grammatical, and fail to do so if it is ungrammatical. We expect them to correct the ungrammatical prompts. Crucially, we should find consistent rates of repetition for existential and predicative constructions. On the other hand, if children have difficulties with the syntax of agreement, we should find repetition match rates to be modulated by syntactic complexity. Children may deviate from adult grammar asymmetrically in the existential constructions, despite being exponed by the same morphological forms as the predicative constructions. The errors should, furthermore, specifically take the form of under-agreement: a preference for copular omission, bare forms over agreeing forms, or default third-singular forms over plural forms.

2.2 Results

De-identified data and code for this and all subsequent experiments are here: https://osf.io/k8nge/?view_only=85ddd414f1e241e9bc022a2f8bdae0dd.

Figure 2 displays the rates of repetition match across canonical and non-canonical trials, by verb agreement form and the semantic number of the associate.

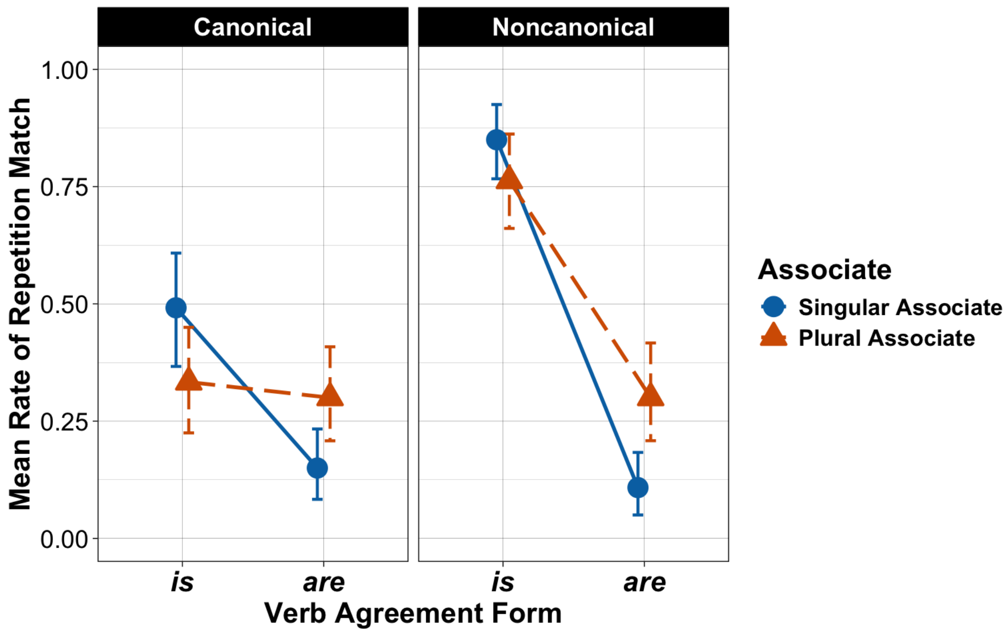


Figure 2: Mean rate of repetition in Experiment 1 as a function of verb agreement (*is* vs. *are*), associate number (Singular Associate: blue circles, solid line; Plural Associate: orange triangles, dashed line), and construction type (Canonical: left panel; Noncanonical: right panel). Error bars are bootstrapped 95% confidence intervals.

Our main analytic question is how repetition rates are influenced by construction (canonical vs. non-canonical), verb agreement form (singular: "is", plural: "are"), and associate number (singular, plural). To explore this, we fit a mixed-effects logistic regression model (using *lmer* package in R) to predict the rate of repetition match as a function of these three factors, with random intercepts for participants.[3] All factors were sum-coded to reveal main effects. The significance of each term (main effects and interactions) was assessed via Type III Wald χ^2 tests using the *car* package.

Participants were significantly more likely to repeat the prompt as given on non-canonical constructions ($\chi^2(1) = 31.56$, $p < .001$). Prompts with plural verb forms ("are") were overall less likely to be repeated ($\chi^2(1) = 150.64$, $p < .001$), though this effect varied by construction (Construction × Verb interaction: $\chi^2(1) = 45.22$, $p < .001$): the difference in repetition rates between singular and plural forms was less pronounced in the canonical construction compared to the non-canonical. This was because the repetition rates were overall low, unexpectedly, with the canonical construction. Verb further interacted with Associate ($\chi^2(1)$ $= 33.57$, $p < .001$), as the probability of repetition increased when the plural verb form oc-

[3]Models with more complex random effects structure failed to converge.

curred with a plural associate, i.e. when the construction would be grammatical for adults. See Table 3 for full model output.

	Estimate	Std. Error	z	p
Intercept	-0.50	0.18	-2.79	0.005
Construction$_{non\text{-}can}$	0.51	0.091	5.62	<.001
Verb$_{are}$	-1.20	0.10	-12.27	<.001
Associate$_{plural}$	0.16	0.09	1.77	0.08
Construction*Verb	-0.62	0.09	-6.73	<.001
Construction*Associate	0.08	0.09	0.87	0.39
Verb*Associate	0.53	0.09	5.79	<.001
Construction*Verb*Associate	0.04	0.089	0.45	0.65

Table 3: Parameter values for fixed effects in mixed-effects regression model of repetition rates; Experiment 1

The natural expectation if there is a late under-agreement stage is a three-way interaction among construction, verb form, and associate number: children should fail to repeat back grammatical plural forms (i.e. those that co-occur with plural associates), but only in the non-canonical construction. Our pre-registered model did not find this interaction (Construction × Verb × Associate interaction: $\chi^2(1) = 0.21$, $p = .65$). Critically, however, this lack of interaction was not due to children failing to display the expected pattern of under-agreement with non-canonical prompts. Indeed, with non-canonical constructions, children overwhelmingly avoided repeating plural forms even when grammatical, and repeated singular forms even when ungrammatical, strongly showing a preference for the default form. The lack of interaction, instead, stemmed from surprisingly low repetition rates with canonical prompts.[4]

[4] We also ran a further exploratory model looking at potential age effects by including Age group as a further fixed effect. Age group was Helmert coded (3-year-olds vs. 4s; younger [3+4] vs. older [5]). Nuanced developmental effects emerged, but none that presents Age as the driving factor in the selective low repetition for non-canonical plurals. We found a significant Construction × Age interaction ($\chi^2(2) = 15.36$, $p < .001$), driven entirely by the 3- vs. 4-year-old contrast ($\beta = 0.58$, z = 3.97, $p < .001$). At age 4, but not 3, there emerged a preference for non-canonical constructions. Whereas 3-year-olds showed low repetition rates across constructions, 4-year-olds showed selectively higher repetition on non-canonical trials. There were two higher-order, three-way interactions involving age: Verb × Associate × Age ($\chi^2(2) = 12.58$, $p = .002$) and Construction × Associate × Age ($\chi^2(2) = 7.66$, $p = .019$). The Verb × Associate × Age effect was again localized to the 3- vs. 4-year-old contrast ($\beta = 0.49$, z = 3.34, $p < .001$). At 4, children more often repeated prompts where verb and associate agreed in number (e.g. plural verb with plural associate), construction-independently. The Construction × Associate × Age interaction was, in contrast, driven by the younger-vs-5-year-old contrast ($\beta = 0.18$, z = 2.50, $p = .012$). There was greater repetition of “are” in non-canonical constructions among the oldest group, but now irrespective of Associate, perhaps suggesting that older kids are more likely to remember and reproduce the sentence faithfully. Critically, nothing here suggests that the hypothesis-relevant selective dispreference for non-canonical plural agreement was primarily driven by Age.

Finally, to better understand the low repetition rates here, we examined children's repair strategies in cases where they failed to repeat the sentence verbatim. Figure 3 illustrates the types of repairs children made. Note that because changes to the associate were so infrequent, both here and in Experiment 2, these repair types were collapsed with the "other" category.

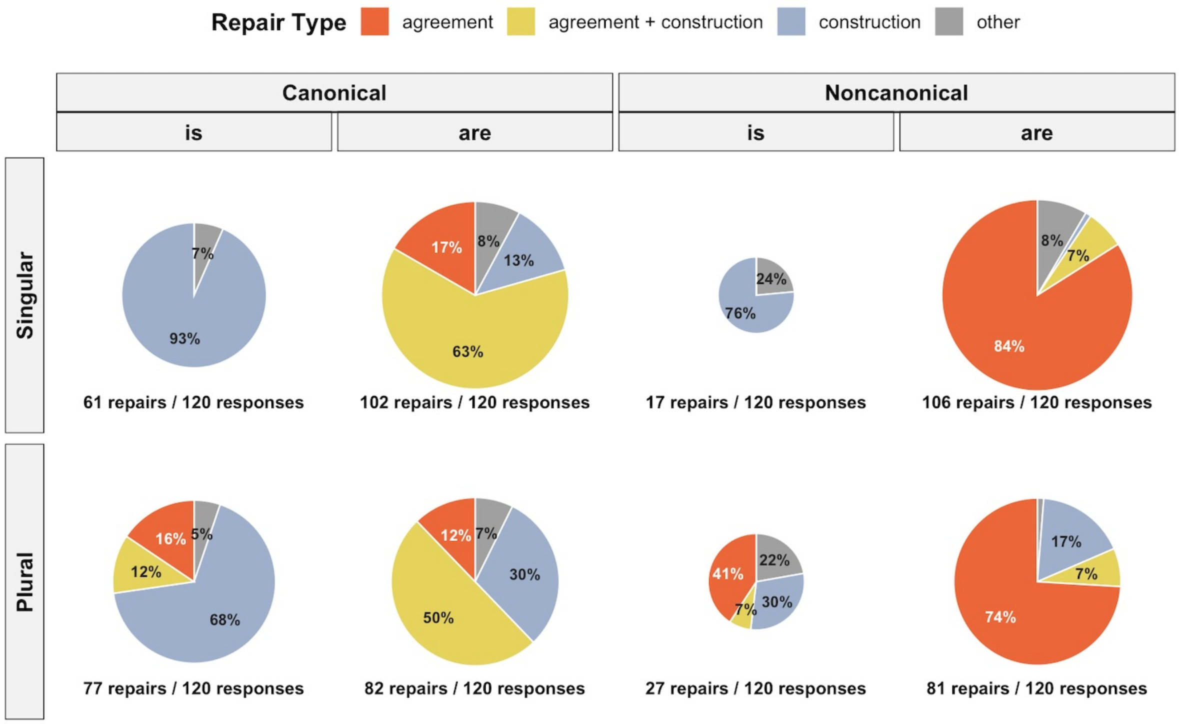


Figure 3: Distribution of repair types among non-repetition responses in Experiment 1, broken down by construction type (Canonical, Noncanonical), associate number (Singular, Plural; rows), and verb agreement form (*is, are*; columns within construction). Each pie represents repairs from a single condition cell; pie area is proportional to the absolute count of repairs per cell (smaller pies mean there were fewer repairs); repair Ns printed beneath each pie.

The left panel of Figure 3 represents repairs on the canonical construction. Interestingly, the dominant change with these sentences was a change in construction. Specifically, children tended to transform the canonical predicative prompts to non-canonical *there*-existential variants. And when the prompt involved a plural verb, they furthermore changed it from "are" to "is," mirroring their behavior with the non-canonical prompts. The right panel shows the pattern of repairs within non-canonical constructions. Notably, the primary mode of repair here consisted of agreement changes, with most involving a switch from "are" to "is". For grammatical cases with a plural associate and plural verb, 60 out of 81 repairs entailed changing "are" to "is". There were some repairs in the opposite direction (from "is" to "are") for ungrammatical cases involving a plural associate and singular verb, though these were less frequent: 11 out of 27 repairs in this condition involved an agreement change.

To statistically probe these differences, we fit an additional mixed-effects logistic model asking whether the rate of agreement-only repairs varied by construction type and verb form (model syntax: `AgreementRepair ~ Construction * Verb + (1|Participant)`). The model revealed a significant interaction between construction type and verb (β=0.47, z=3.30, $p < 0.001$), with agreement-only repairs being more prevalent with plural verbs, but only in non-canonical constructions. In other words, while children consistently showed low repetition rates for plural forms, the underlying reasons differed. In non-canonical constructions, the reduced repetition rate stemmed from a clear preference for singular agreement. By contrast, in canonical constructions, it was driven by a preference for an entirely different construction.

2.3 Discussion

In Experiment 1, we used *there*-existentials as our case study to probe children's command of non-canonical agreement. Our results reveal a notable point of divergence from adult grammar: a preference for singular agreement. When the prompt they were given was singular, children repeated it as is whether grammatical (12-a) or ungrammatical (12-c). They correct ungrammatical cases of plural agreement with a singular associate (12-b). But strikingly, they corrected grammatical prompts where the verb agreed in the plural with a plural associate (12-d).

(12) a. Is there a toy in the box? → Is there a toy in the box?
b. Are there a toy in the box? → Is there a toy in the box?
c. Is there some toys in the box? → Is there some toys in the box?
d. Are there some toys in the box? → Is there some toys in the box?

We propose that these findings can be straightforwardly understood as a case of *failure to agree*. In child existential constructions, as in their adult counterparts, the probe on T first targets the expletive and moves it to the subject position. However, unlike in the adult grammar, the second instance of Agree fails to occur, leaving the ϕ-features on T unvalued. These unvalued ϕ-features are then spelled out by a morphological default, which, in English, happens to be the 3Sg form.

Our main claim, thus, is that preschoolers' errors in non-canonical constructions are syntactically driven, incomplete mastery of two-step Agree dependencies or difficulty implementing those dependencies online. Apparent singular agreement arises from agreement failure, with 3Sg verbal morphology being a default form. In other words, what we observe here is a late under-agreement stage.

It is important to note here that our analysis of child grammar in *there*-existentials does not represent a radical departure from adult grammar. In fact, it has long been observed that certain dialects of English allow singular agreement with plural associates in existential constructions (Emonds (1985), Safir (1985), Sobin (1997), Sobin (2014), Smallwood

(1997), Schütze (1999)). While some dialects condition this on the contracted form of the copula (*there's*), for a substantive minority of speakers, singular agreement is grammatical regardless of contraction (see Smallwood (1997) for elicited production data). Schütze (1999) provides an analysis of this phenomenon in adult English that closely resembles our account for child English: when agreement with the subject is not possible, some speakers default to a non-agreeing option and insert a morphological default in place of the regular agreement morphology.

One intriguing possibility is that agreement challenges during childhood may have been the catalyst for the emergence and eventual solidification of the non-agreeing option in these speakers. At the same time, the fact that singular agreement with plural associates is grammatical for at least some adult speakers leads us to ask: Could children's under-agreement in existential constructions be driven by an over-representation of non-agreeing forms in adult speech, in particular child-directed speech? In other words, are children simply reproducing what they hear? While we do not have access to the language environment of the specific children in our study, here we turn to two types of proxy measures. First, we conduct an analysis of *there*-constructions in child-directed speech using existing corpora in CHILDES (MacWhinney, 2000). Second, we examine how adults evaluate the kinds of sentences used in this experiment, with both matching and mismatching agreement. We turn to these next.

Before moving on, let us say a word about the low rates of repetition we observe for the canonical constructions, as neither a purely syntactic nor a purely morphological account predicts this. We suspect that our experimental setting itself may have encouraged children's robust use of existentials, even when prompted with canonical predicative sentences. Sentences containing indefinites can be presuppositional when they appear in sentences of the form 'Some toys are in the box': they take for granted the existence of the toys (Diesing, 1992; von Fintel, 1998). But this presupposition is not licensed in our experimental discourse context, which presumes ignorance about the contents of the box. There is a long line of research shows that children are highly sensitive to such subtle infelicities in experimental settings, and their linguistic behavior can reflect this sensitivity (Crain & Thornton 1998). We believe that this is precisely why we find low rates of repetition with canonical constructions in this experiment. They are somewhat odd in the context, children are exquisitely sensitive to pragmatic ill-formedness, and the *there*-construction is a type of repair that aims to fix the ill-formedness.

3 Study 1B: Corpus follow-up

To better understand the linguistic input that children receive with *there*-existentials, we turn now to an examination of child-directed speech.

As mentioned earlier, there is one existing corpus study that examined the agreement forms present in children's input for there-existentials: Lukyanenko & Miller (2023). The authors looked at longitudinal data from two corpora in CHILDES (MacWhinney, 2000), Sarah (ages 2;3-5;1 [Brown (1973)]) and Nina (ages 1;11-3;3 [Suppes (1974)]), as well as cross-sectional data from 100 parent-child dyads who read a picture-book. Their main finding was that at least some caregivers produce under-agreeing forms in *there*-constructions. However, two limitations in their study lead us to probe this issue further. First, their primary analysis classifies both singular agreement and contracted *-s* with plural subjects as cases of non-agreement (though they are careful in acknowledging that high rates of contraction might weaken their conclusions). This collapsing is problematic because there is no consensus on how to analyze the contracted form in *there*-constructions: Is it a phonologically reduced variant of *is* (Labov, 1969), an allomorph that can realize both singular and plural agreement in the syntax (Safir, 1985), or an unanalyzable piece of a fossilized chunk (Rupp & Britain, 2019)? A further complication stems from the fact that their analysis did not differentiate between declaratives and questions. This was critical to our design: contraction is not possible in questions, making them a better test environment.

In our analysis, we focus specifically on patterns of agreement in *there*-existential constructions, distinguishing between questions and declaratives.

3.1 Corpus

We make use of the Providence Corpus available on CHILDES (Demuth et al. (2006); MacWhinney (2000)), which contains data from six children (three boys and three girls) and consists of hour-long, naturalistic recordings of mothers interacting with their children at home during everyday activities such as playtime, meals, and reading. [5] Collected between 2002 and 2005 in Providence, Rhode Island, these recordings began when the children first started to produce words (between 11 and 16 months) and continued every two weeks for 2–3 years, with most data covering ages 1;0 to 3;6. Both child-directed speech (from caregiver to child) and child speech are included. Audio recordings were transcribed using the CHAT (Codes for the Human Analysis of Transcripts) format, the standard system used in the CHILDES database. This corpus was chosen for its substantial data volume (364 hours of speech) and the inclusion of video data, which allows annotators to reference the physical context if the audio or transcriptions are insufficient. Table 4 provides details on the children's ages and the total recordings per child.[6]

3.2 Procedure

Using the CLAN program, we extracted all instances of *there* with the copula, employing the COMBO command to search for specific text strings. Since we were interested in the child's input, we restricted our search to the MOT tier (using the *+t* switch). We extracted

[5] Although other speakers (such as fathers, grandparents, or other relatives) surface in some of the files, only mothers are present consistently across all files and children and provide the bulk of the adult data. We thus focus on this dyad here.

[6] One child, Ethan, was diagnosed with Asperger's Syndrome at age 5, but since we focus only on caregiver speech and his diagnosis occurred two years after recording ended, we opted not to exclude his files.

Name	Age range	Sessions
Alex	1;04.28-3;05.16	51
Ethan	0;11.04-2;11.01	50
Lily	1;01.02-4;00.02	80
Naima	0;11.27-3;10.10	88
Violet	1;02.00-3;11.24	51
William	1;04.12 - 3;04.18	44

Table 4: Age ranges and number of recording sessions per child in the Providence Corpus

all occurrences of *there is*, *there are*, *is there*, *are there*, and the contraction *there's* (with the +*s* switch, e.g., +s"there^is"). This process yielded a total of 3,986 caregiver utterances. We manually reviewed these to ensure they were genuine *there*-existentials, and not sentences involving locative *there*. This yielded 3,876 utterances for further annotation after exclusions.

The utterances were then annotated for Sentence Type (declarative, question), Associate (singular, plural, or other/unspecified), and Agreement form ("is," "are," contracted "-'s"). The Associate was categorized as "other/unspecified" in instances where the utterance was incomplete, those involving NP ellipsis (e.g., "there was some Δ."), or when the Associate was a mass noun (e.g., "there was some milk").[7] As mentioned above, the treatment of the contracted copula -*'s* varies across theories, so we code it separately as its own category.

3.3 Results and discussion

We excluded utterances where the Associate was coded as "other/unspecified," yielding 3,783 utterances after exclusion for analysis. Figure 4 shows the relative proportions of the three different copular forms used by caregivers in existential declaratives and questions when the Associate was singular versus plural. Table 5 provides the raw counts for each combination of Sentence Type, Associate, and Agreement form.

To analyze these data, we fit a Poisson generalized linear mixed-effects model predicting utterance counts from Agreement Type ("is", "are", "-s"), Sentence Type (declarative, question), and Associate (singular, plural), with all factors sum-coded. The model included random intercepts for Child to account for inter-corpus variability. Observations for which the count was 0 were dropped.

The model revealed that while plural forms were less frequent overall than singular "is" (β=-2.19, z=-5.37, p<.001), this effect interacted with Associate (β=2.34, z=6.88, p<.001). This interaction was driven by the fact that with plural associates, plural agreement forms were more frequent. Frequency of contracted forms also interacted with Associate, with plural associates yielding fewer contracted forms compared to the singular "is" (β=-1.22, z=-6.20, p<.001). The model did not detect any main effects of sentence type. However,

[7] We excluded mass nouns because these are not part of the same paradigm. Even though they routinely take singular morphology, they are not *semantically* singular.

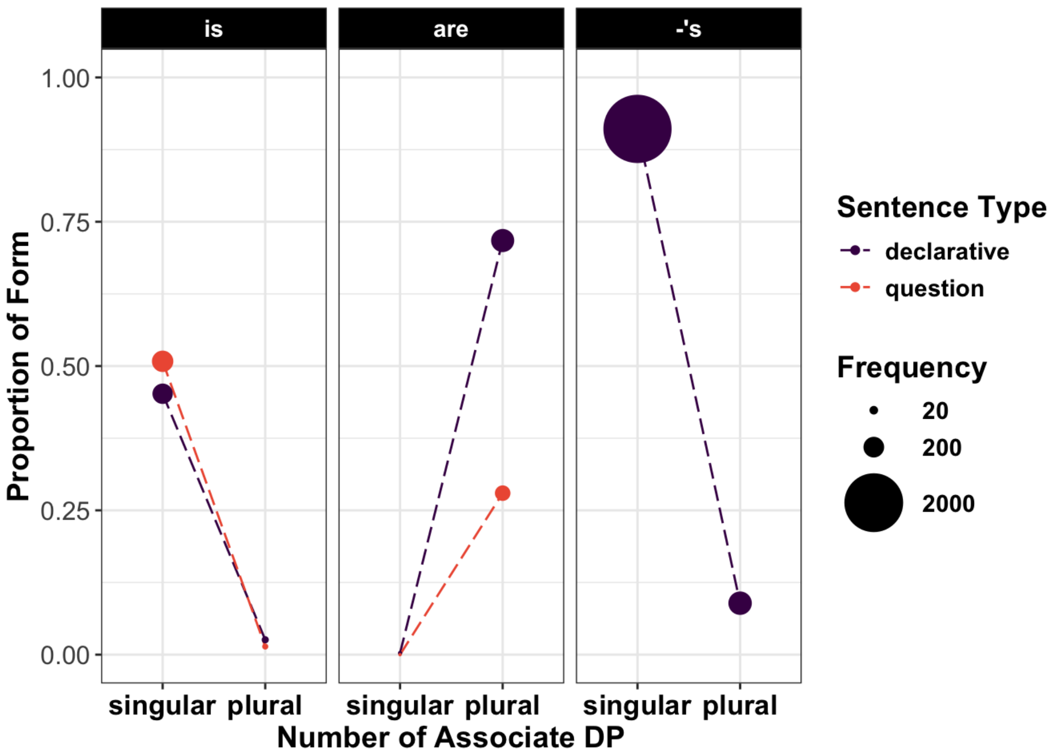


Figure 4: Relative proportions of each verb agreement form by sentence type and associate. Point size indicates the number of contributing utterances.

Sentence Type	Associate	Agreement	Count
Declarative	Singular	*is*	193
		are	1
		-*'s*	2728
	Plural	*is*	11
		are	259
		-*'s*	267
Question	Singular	*is*	217
	Plural	*is*	6
		are	101

Table 5: Distribution of Agreement Forms by Sentence Type and Associate

it is important to note that the key point of divergence — the availability of contracted -*'s* —could not be assessed, as its count was zero in questions.

Overall, these data suggest that children are not exposed to a substantial proportion of uncontracted *is* with plural Associates. Although singular Associates are more frequent overall, children do hear plural Associates with the expected plural agreement forms. Specifically, when the Associate is plural, caregivers use the plural form *are* (48% of the time) or contracted *-'s* (50% of the time) in declarative sentences. In questions, where contraction is not an option, caregivers use plural agreement 96% of the time when the Associate is plural.

Thus, while children hear some instances where singular agreement is applied to plurals, such cases are only a minority. Importantly, they hear plenty of plural Associates with plural agreement forms. We take this as evidence that input alone cannot fully explain children's preference for singular agreement.

4 Study 1C: Adult control study

As a second proxy measure, we conducted an adult control study using the same (or closely matched) stimuli that children had seen.

4.1 Participants

The analysis includes data from N = 50 adults, recruited online via Prolific. All participants self-identified as native speakers of English and had US IP addresses. An additional 10 participants were tested but excluded due to low accuracy (<70%) on catch trials.

4.2 Materials, Design, and Procedure

Because the linking assumptions underlying the elicited production task do not hold for adults the same way as with children, the adult study used a forced-choice task. We cross construction type (canonical vs. non-canonical) with associate number (singular vs. plural) in a 2x2 within-subjects design. Each participant completed 32 critical items (8 per condition), presented in randomized order via Qualtrics. On each trial, participants chose between two versions of a sentence, one agreement-matching and one mismatching, or indicated that both were acceptable. See Figure 5. If adults behaved in a manner parallel to the children, they should opt for the agreement-mismatching option in non-canonical sentences with plural associates, but prefer matching agreement elsewhere.

Catch trials were included throughout, consisting of copular constructions (e.g., John is a doctor), with mismatching versions created by omitting the copula (e.g., John a doctor). These constructions were included both as attention checks and to exclude speakers of dialects whose treatment of existentials differs substantially from the variety under investigation. Some dialects of English permit copular omission in cases like our catch trials (see e.g., Green, 2002), and these dialects often also differ in the syntax of existentials in ways that directly affect agreement patterns — for example, by allowing expletive *it* or *they* in place of *there* (Green, 2002). Given this, selection of the option without the copula, or selection of 'both OK' were counted as 'errors' and figured into our exclusion calculations.

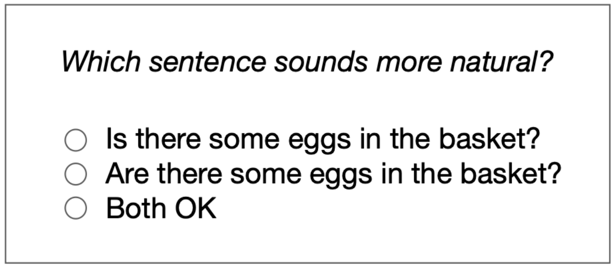


Figure 5: Trial structure for the adult forced choice task.

4.3 Results and discussion

Figure 6 shows the distribution of adults' selections across conditions. Adults overwhelmingly chose agreement matches, though other responses were attested. We asked two questions: (1) do adults' choices with respect to agreement vary by construction type, and is this modulated by associate number? (2) do adults disprefer plural agreement in non-canonical constructions, as we observed in children?

To address the first question, we fit a mixed-effects logistic regression predicting choice of Match Selection from Construction (canonical, non-canonical) and Associate Number (singular, plural), both sum-coded, with random intercepts for Participant. The model revealed a significant interaction between construction and associate (β = –0.39, SE = 0.11, z = –3.63, $p < .001$). This interaction was driven by lower rates of match responses in non-canonical constructions specifically when the associate was plural.

To test whether this pattern reflected a broader dispreference for plural agreement, we conducted an additional, more flexible analysis. This time choices of Match or Both were grouped together and compared against Mismatch. This allowed us to probe when participants *preferred* the singular agreement option, as opposed to considering it acceptable alongside the plural (i.e. selecting Both). This model revealed a robust main effect of associate number (β = –0.77, SE = 0.15, z = –5.19, $p < .001$), such that plural associates elicited more Mismatch selection overall. Crucially, however, there was no interaction with construction type.

On the one hand, our findings suggest that adults are somewhat more flexible in their agreement judgments with non-canonical constructions, occasionally allowing singular agreement with plural associates in these contexts. On the other hand, this flexibility differs from children's behavior both qualitatively and quantitatively. Children in our study actively repaired plural agreement, even when it was grammatical. Adults do not do this: plural agreement with plural associates was always accepted as an option. At the same time, the fact that some adults also accepted singular agreement with plural associates suggests that there may remain an advantage, even in the adult grammar, to selecting the underagreeing variant. We return to this point in the discussion.

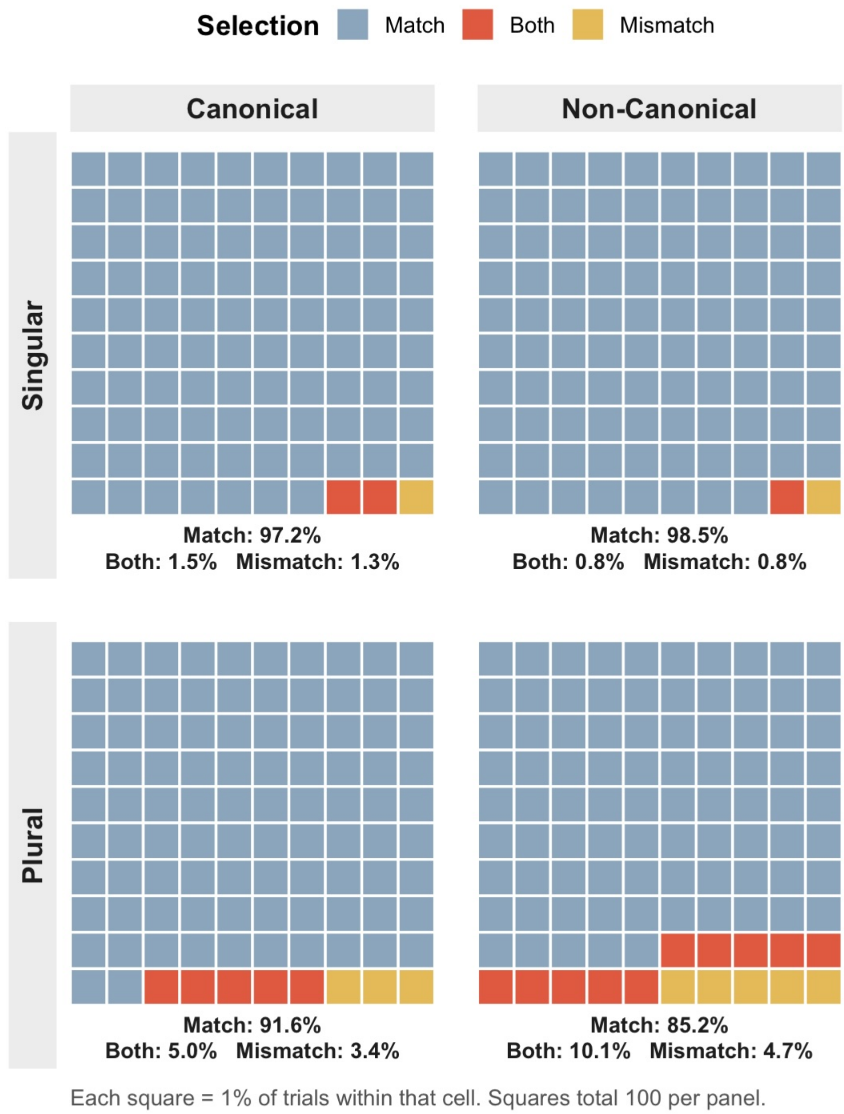


Figure 6: Distribution of adult selections in Experiment 1C (Existentials) by Construction (Canonical, Non-Canonical) and Associate Number (Singular, Plural). Each panel shows the response distribution for one trial type as a 10 × 10 grid of 100 squares, where each square corresponds to one percent of the trials in that cell. Squares are colored by Selection, and the percentage of each response category is printed beneath each panel.

5 Interim summary

With *there*-existentials, children reproduced ungrammatical sentences with singular agreement and plural associates, and repaired grammatical sentences with plural agreement and plural associates. This pattern was not robustly found either in child-directed speech, or in adult language generally. We argued that children's non-adult singular preference in non-canonical agreement structures reveals a late under-agreement stage. Specifically, we suggested that their 3rd singular forms result from a failure to carry out the second cycle of Agree necessary for valuing the ϕ−features on T, and realizing the unvalued features with a default. This, we suggested, can be seen as continuous with younger children's difficulties with agreement in canonical configurations.

However, this conclusion can be challenged in two ways. First, the canonical constructions we included as controls — meant to demonstrate that children past age 3 are adult-like with agreement in syntactically simpler cases — did not display the expected patterns. This is not only inconsistent with our expectations, but also conflicts with prior findings, which indicate that preschoolers do not exhibit under-agreement in canonical constructions. A closer examination of children's repair patterns revealed that their performance in our task was partly driven by a strong preference for existential constructions. Nevertheless, we need to establish a reliable baseline before drawing any firm conclusions about agreement in non-canonical cases.

Second, there is a plausible, non-syntactic account that we have not yet ruled out. This alternative is suggested by the existence of other expletive constructions in English that reliably trigger canonical agreement. Consider the third person singular expletive *it*, which is thought to have the same ϕ-featural make-up as ordinary third-person pronouns (Longenbaugh, 2019). In sentences where expletive *it* serves as the subject, unlike those with expletive *there* subjects, the verb uniformly agrees in third-person singular with the expletive subject, rather than the associate; see (13).

(13) a. It {is/*are} equally likely for it to rain and for it to snow.
b. For it to rain and for it to snow {??is/are} equally likely.

It is possible that children initially expect formal uniformity across different expletives, and assume that both *it* and *there* trigger canonical singular agreement. If so, the errors we saw with *there*-existential constructions could still have a morphological explanation: children have yet to learn properties of a specific morpheme, namely expletive *there*. When they produce singular agreement with plural associates, that would then be canonical agreement, rather than an instance of agreement failure. Third singular verbal morphology, in turn, is not a default, but the output of a successful Agree relation.

To test this alternative account, in Experiment 2 we extend our investigation to a different non-canonical agreement structure: locative inversion. In these constructions, the verb agrees with a post-verbal noun phrase, but, crucially, the subject is not an expletive; instead, it's a locative prepositional phrase. This allows us to explore children's understanding of non-canonical agreement in a context without expletives. If the issue is specific to expletives, we should see adult-like agreement patterns with locative inversion. If, on the other, the problem is genuinely syntactic, it should generalize across non-canonical agreement configurations and we should replicate the findings from Experiment 1.

6 Experiment 2

Experiment 2 examines locative inversion constructions like the examples in (14):

(14) a. In the box is a cookie.
     b. In the box are some cookies.

(15) a. A cookie is in the box.
     b. Some cookies are in the box.

Unlike their non-inverted counterparts (15), locative inversion constructions exhibit a distinctive agreement pattern: the verb morphology co-varies with the postverbal nominal, and the locative PP surfaces to the left of the verb. Following Bresnan (1994), Culicover & Levine (2001) and Doggett (2004), we assume that these properties result from the PP — rather than the DP, like in (15)— raising to subject position. Since PPs lack ϕ-features, T probes a second time, finding the necessary ϕ-features on the postverbal DP. The second step of Agree is thus analogous to what we saw with *there*-existentials.

It should be noted that the syntax of locative inversion is far from settled, with substantial debate over whether the fronted PP genuinely occupies the subject position. There are two main schools of thought: one posits that the PP itself raises to Spec-TP, and then subsequently moves to a topic, while the other suggests that the inverted PP moves directly to a topic position, with a null expletive like *there* occupying Spec, TP. We find the arguments from Bresnan (1994) for the subjecthood of the locative persuasive. She presents a number of diagnostic tests that distinguish the PPs in locative inversion from topicalized PPs in sentences with an expletive *there*, and demonstrates that the PP triggers the *that*-trace effect, fails to trigger *do*-support, can undergo subject-to-subject raising, parallelism for ATB extraction, etc., only in the locative inversion construction. (16) illustrates the *do*-support diagnostic.

(16) a. Which picture {hung/*did hang} on the wall?
     b. On which wall {hung/*did hang} a portrait of the artist?
     c. On which wall {*hung there/did there hang} a portrait of the artist?

Other contested points, such as the mechanics of how the PP raises past the DP, are less central to our argument. Some approaches (e.g., Anagnostopoulou (2003); Bruening (2001); Ura (2000)) propose that T can Agree with the PP because the PP and the DP are equidistant, neither being closer to T than the other. Others suggest an intermediate movement step within the *v*P, potentially driven by additional A-bar features on the PP, where it raises above the DP. Whichever approach is correct, both entail a division of T's needs across two phrases, yielding a more complex agreement configuration than that of the non-inverted variant.

6.1 Methods

This experiment was not separately pre-registered, but had the same sample size and followed the same analysis plan as Experiment 1.

6.1.1 Participants

We analyze results from a separate sample of 60 children (Mean Age = 4;7, Range = 3;0-5;11), whose dominant language (at least 60% of use) was English. Participants were recruited from preschools and museums in the MASKED area. We did not collect other demographic information for the study. An additional 26 children were tested but excluded for failure to complete the study (15), language delay (1), English not being the dominant language (3), parental interference (3), or technical or transcription issues (4).[8]

6.1.2 Design, Materials and Procedure

Design, materials and procedures were only minimally different from Experiment 1. We again used a 2x2x2 design, crossing Construction (canonical predicative vs. non-canonical locative inversion), Associate Number (singular vs. plural) and Verb Agreement (is vs. are), yielding 8 within-subject conditions, half of them ungrammatical. See Table 6. The form of the prompt was changed to incorporate locative inversion sentences as the non-canonical construction of interest. Furthermore, we used the declarative forms for both canonical and non-canonical constructions, as polar questions are blocked with locative inversion (**Is in the box a cookie?*). Accordingly, participants were told that they were going to make guesses about what is in Shy Giraffe's containers, and that he would reveal the contents only if both parties make the same guess.

	Verb Agreement	**Associate Number**	
		Singular	Plural
Non-canonical	is	In the box is a toy	In the box is some toys
	are	In the box are a toy	In the box are some toys
Canonical	is	A toy is in the box	Some toys is in the box
	are	A toy are in the box	Some toys are in the box

Table 6: Trial types, Experiment 2

[8]The exclusion rate is much higher for this experiment compared to Experiment 1 because of a change in testing setting from preschools to a busy children's museum.

6.2 Results

As before, our main dependent measure is whether or not children repeated the prompt as given. Figure 7 displays the rates of repetition match across canonical and non-canonical trials, by verb agreement form and the semantic number of the associate.

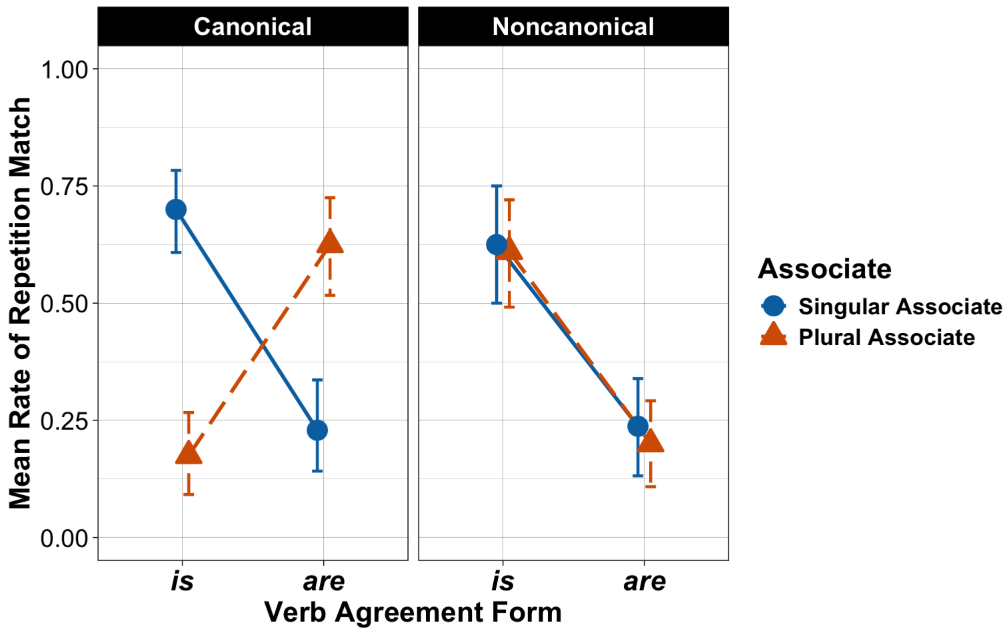


Figure 7: Mean rate of repetition in Experiment 2 as a function of verb agreement (*is* vs. *are*), associate number (Singular Associate: blue circles, solid line; Plural Associate: orange triangles, dashed line), and construction type (Canonical: left panel; Noncanonical: right panel). Error bars are bootstrapped 95% confidence intervals.

We analyzed repetition rates using a mixed-effects logistic regression model with repetition match as our response variable, and construction (canonical vs. non-canonical), verb agreement (singular: "is", plural: "are"), and associate number (singular, plural) as fixed effects, all sum-coded. The model included random intercepts for participants. There was a significant three-way interaction (Construction × Verb × Associate: $\chi^2 = 69.34$, $p < .001$). Participants were less likely to repeat plural verb forms ("are") with plural associates, but only with the non-canonical (locative inversion) constructions. Table 7 displays the full model output.[9]

[9] As in Experiment 1, we also looked at age effects, by including Age group (3 levels; Helmert-coded as before) was added as a fixed effect. Once again, we found developmental differences that affected some components of repetition behavior, but not the critical effect. There was a main effect of Age ($\chi^2(2) = 8.15$, p = .02), driven primarily by an overall increase in repetition rate between ages 3 and 4 ($\beta = 0.75$, SE = 0.31, $z = 2.426$, $p = .015$). There was a significant Verb × Age interaction ($\chi^2(2) = 18.23$, $p < .001$), this time driven by the younger (3+4) vs. older (5) contrast ($\beta = 0.24$, SE = 0.06, $z = 3.7$, $p < .001$). The oldest group was more likely to repeat "are". No higher-order effects of Age were found, so once again age doesn't seem to be

	Estimate	Std. Error	z	p
Intercept	-0.57	0.25	-2.29	0.02
Construction$_{non\text{-}can}$	-0.04	0.09	-0.42	0.68
Verb$_{are}$	-0.69	0.09	-7.23	<.001
Associate$_{plural}$	-0.17	0.09	-1.90	0.06
Construction*Verb	-0.63	0.09	-6.66	<.001
Construction*Associate	0.06	0.09	0.67	0.50
Verb*Associate	0.72	0.10	7.58	<.001
Construction*Verb*Associate	-0.80	0.10	-8.33	<.001

Table 7: Parameter values for fixed effects in mixed-effects regression model of repetition rates; Experiment 2

We also explored children's repair strategies when they fail to repeat the prompt; see Figure 8. Of 531 total repairs, repairs involving agreement changes were the most frequent (183).

A mixed-effects logistic regression model predicting agreement-only repairs as a function of construction type and verb form revealed significant effects of construction (β=-0.78, z=-4.71, $p < 0.001$), verb form (β=0.53, z=3.26, $p = 0.001$) and an interaction between the two factors (β=0.69, z=4.09, $p < 0.001$). Participants were less likely to opt for agreement-only repairs in non-canonical constructions compared to canonical ones (74 vs. 109 instances). There was also an overall greater tendency to change just agreement with plural verb forms than singular ones (123 vs. 60). However, in non-canonical constructions, this difference between singular and plural verb agreement was smaller, driven by the fact that agreement changes in these cases were often also coupled with a construction change (i.e., changes to the syntax of any kind; see Figure 8 right). Virtually all of these involved a switch to the *there*-existential, either replacing the PP altogether, or between the PP and the copula. The higher rate of construction changes raise the possibility that fewer children may be familiar with locative inversion constructions to begin with, a possibility we return to later.

6.3 Adult controls

As we did with *there*-existentials (see §4), we ran an adult control study using materials closely matched to what children saw in Experiment 2.

6.3.1 Participants

We recruited a new sample of native English-speaking adults via Prolific. Fifty-three participants were included in the analysis. An additional seven participants completed the

driving the selective under-agreement pattern.

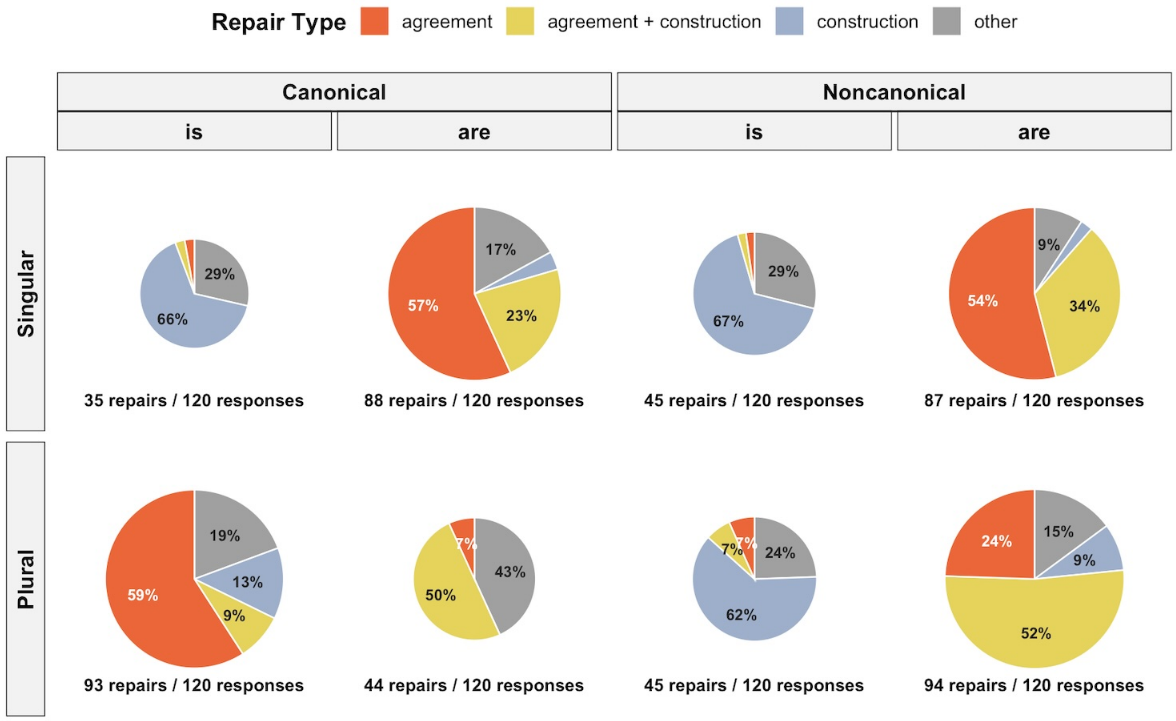


Figure 8: Distribution of repair types among non-repetition responses in Experiment 2, broken down by construction type (Canonical, Noncanonical), associate number (Singular, Plural; rows), and verb agreement form (*is, are*; columns within construction). Each pie represents repairs from a single condition cell; pie area is proportional to the absolute count of repairs per cell (smaller pies mean there were fewer repairs); repair Ns printed beneath each pie.

study but were excluded due to low performance (< 70% Accuracy) on catch trials.

6.3.2 Materials, Design and Procedure

The design, procedure, and criteria for inclusion/exclusion were the same as those used in the adult control study for *there*-existentials (Section 4). Adults completed a forced-choice task in a 2 × 2 design crossing construction type (canonical vs. non-canonical) with associate number (singular vs. plural). Each participant saw 32 critical items (8 per condition) and 16 catch items. The critical items involved sentences either identical or highly similar to the ones children saw in Experiment 2. The catch items were the same as in Experiment 1C. On each trial, participants chose between two versions of a sentence, one matching in agreement, one mismatching, or indicated that both were acceptable.

6.3.3 Results and discussion

Participants choices across conditions are visualized in Figure 9.

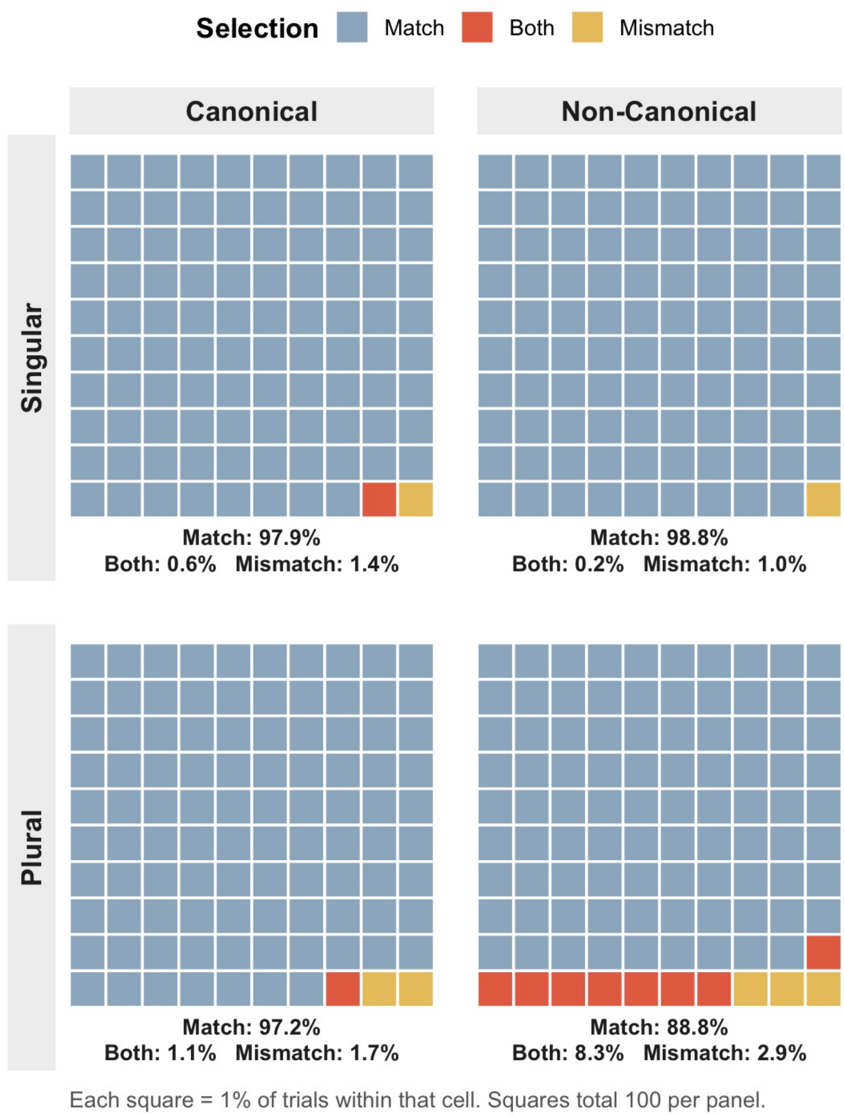


Figure 9: Distribution of Selections in the Locatives Adult Control Experiment, split by Construction (Canonical, Non-Canonical) and Associate Number (Singular, Plural) . Each panel shows the response distribution for one trial type as a 10 × 10 grid of 100 squares, where each square corresponds to one percent of the trials in that cell. Squares are colored by Selection, and the percentage of each response category is printed beneath each panel.

Responses closely mirrored the patterns observed in Study 1C. Adults overwhelmingly preferred agreement matches across all conditions. Our analytical approach is identical to that in Study 1C. We fit two mixed-effects models, one predicting the rate of “Match” responses and another using a broader dependent variable, “Match or Both”

The first model revealed a significant interaction between Construction and Associate number (β = -0.58, SE = 0.12, z = -4.83, $p < 0.001$), reflecting lower match rates for non-canonical sentences when the associate was plural. The second model showed only a main effect of plural associates (β = -0.35, SE = 0.15, z = -2.35, p = 0.02), indicating that mismatches were generally more frequent with plural associates, but this effect did not depend on construction type.

Thus, as with the existentials, adults showed slightly more agreement flexibility in non-canonical constructions. But there is no strong preference for singular over plural agreement with plural associates. Grammatically correct plural agreement is always accepted.

6.4 Discussion

Our findings in Experiment 2 are consistent with children going through a late under-agreement stage. We found that in canonical constructions, children’s repetition and repairs were modulated by grammaticality: children preferred singular agreement with singular associates, and plural agreement with plural associates. But in non-canonical constructions, there was a marked preference for singular agreement. Children repeated locative inversion sentences with a plural associate and singular verb, and repaired cases where a plural associate was paired (in line with the target grammar) with a plural verb.

Critically, what we find with locative inversion constructions is similar to what we saw in Experiment 1 with *there*-existentials. This helps rule out the possibility that children’s singular preference in those constructions was due to them treating the expletive as triggering *canonical* 3Sg agreement. But the problem persists even in the absence of an expletive. To the extent that what *there*-existentials and locative inversion constructions share is a more complex agreement profile, we can conclude that the problem is about non-canonical agreement broadly.

Of course, even if in adult grammar, the two constructions are syntactically similar, that does not mean children treat the two constructions as the same at the outset. Locative inversion differs from existentials in several ways. First of all, they simply ‘look’ different, making the connection based on surface evidence harder. Second, locative inversion is infrequent in adult language, making it likely rare in child input as well. As a result, many children may not know the construction well enough, and might even misanalyze it as sharing the same agreement profile as existentials. Indeed, our results themselves point to lower proficiency with these constructions. Repetition rates for non-canonical constructions, even for grammatical sentences with singular associates, were lower here than in Experiment 1. Whereas we found a small effect of grammaticality on plural repetition rates in Experiment 1, suggesting emergent mastery of postverbal agreement, we failed to find a parallel effect in Experiment 2. And unlike Experiment 1, where children’s repairs to existentials rarely involved changing the construction altogether, construction changes were more common in Experiment 2. Together, these patterns are consistent with the possibility some of these children simply do not have locative inversion as part of their grammar. This limits the strength of our conclusions in trying to account for the full pattern of errors and repairs. After all it is possible that some of the repetition failures in the locative inversion construction are due to infrequency, rather than complexity. Still, among the children who do attempt to repeat the construction, we do find selective agreement changes, a pattern that is consistent with children initially adopting an under-agreeing analysis rather than a post-verbal agreement analysis. Furthermore, the most likely candidate for what children could misanalyze in such infrequent constructions is their agreement profile, and the reason for that is its increased complexity. Thus, the parallels we aim to draw across the two constructions remain informative, even if the the additional challenges of learning locative inversion mean they might not be fully comparable.

7 General discussion

7.1 Summary of findings

Our starting point was a developmental puzzle: why do toddler languages show less agreement than the corresponding adult variants? The key debate concerned whether these child-adult differences had syntactic roots, or if they reflect morphological knowledge or performance. To probe this, we turned to an older population — preschoolers — and an understudied empirical domain: non-canonical agreement. Non-canonical agreement constructions let us manipulate syntax independently of morphology. They rely on the same morphological forms as canonical constructions, which preschoolers typically already know and use, but have a more complex agreement profile. Our findings reveal that in canonical constructions, child and adult English align: both populations prefer verb forms that agree with the subject (e.g., "Some cookies are in the box"). However, in non-canonical constructions, their grammars diverge. In adult English, the verb morphology tracks the post-verbal nominal (e.g., "Are there some cookies in the box?" or "In the box are some cookies"). By contrast, children show a singular preference (e.g., "Is there some cookies in the box?" or "In the box is some cookies"). We argued that this is a case of under-agreement, where children do not carry out the second Agree step required for agreement with the post-verbal nominal, instead realizing the unvalued features with default 3rd singular morphology.

Thus, with increased syntactic complexity, there is a re-emergence of under-agreement in a population that has otherwise mastered agreement and has no issues with the necessary morphology. Children at this stage of development can manage single-step dependencies, but have trouble once you go beyond that. The late under-agreement stage could thus be seen as continuous with an earlier developmental stage, where even simpler dependencies posed a challenge.

It is worth emphasizing that our approach to the original puzzle is indirect by design. This allowed us to side-step the inherent difficulties in further exploring toddlers, where disentangling syntactic and morphological factors became increasingly hard. But the indirectness of our approach also limits our conclusions. Here we are not observing the 'classical' under-agreement phenomenon, but something whose resemblance to the original depends on a variety of theoretical assumptions. Yet the fact that similar errors resurface in an older population, and specifically under conditions of increased syntactic complexity, strengthens the case that early and late under-agreement are connected, and reflect the same underlying factor: either a protracted maturation of the syntactic machinery for agreement itself, or a capacity that gets taxed by such syntactic complexity.

Indeed, our adult findings may represent a weaker but still detectable continuation of the same problem. Even adult speakers may find two-step agreement configurations more difficult, resulting in a modestly increased tolerance for singular agreement with plural postverbal subjects, a pattern also consistent with crosslinguistic observations that agreement with postverbal subjects is often less robust. We return to this latter point in §7.3.

7.2 Implications for theories of 'classical' under-agreement

Assuming we accept the connection between early and late under-agreement, our findings lend support to syntactic accounts of early under-agreement. Could non-syntactic accounts explain these data? Emergentist accounts have no straightforward way to do so. Children clearly use the plural form of the copula correctly in canonical constructions, so their errors cannot be due to a general failure to learn the relevant morphology. Nor can they be explained as rote learning of chunks like *there is*, since we see the same errors in locative inversion, where no such fixed chunk is readily available.

On a first look, morphological performance accounts, such as Phillips's (2010), fare no better than the emergentist ones. Recall that on this view, children have the requisite syntactic and morphological knowledge, but may have difficulties retrieving the right forms when structure-building. Why should syntactic complexity exacerbate issues in a competence thought to be independent: morphological retrieval?

But on a closer look, it is not impossible to revise a performance model such that it can accommodate these results. For instance one might assume that there exists a critical capacity, (e.g. working memory) that gets taxed and (potentially bottlenecked) by demands both on morphological retrieval and syntactic complexity. For this shared system, then, the retrieval problems would be exacerbated in constructions with multiple dependencies. On such a hybrid picture, syntactic difficulties would amplify morphological performance issues.

There is reason to think this hybrid view is on the right track for the classical under-agreement stage. In those earlier stages, where even canonical constructions pose difficulty, children's errors with copular constructions look a bit different than what we find here, as younger children often omit the copula altogether (Becker 2004). Indeed non-retrieval might be a more stereotypical solution to retrieval problems than systematic mis-retrieval, which is what the older children in our studies seem to be doing. They no longer omit the copula, but opt for singular forms. This developmental difference would be neatly explained if the syntactic difficulties (i.e. difficulties with agree dependencies) would remain even after other performance problems (i.e. retrieval troubles) are resolved.

Future work could more directly tease apart a purely syntactic view from a hybrid view, by, for example, contrasting the past tense forms of the copula *was* and *were* (e.g., *There were some toys in the box*). Since neither form is a morphological default (Bjorkman, 2011; Rudnev, 2024), we would not expect either to be the form children fall back on when facing retrieval problems. But, if they have difficulties specifically with the second step of agreement, we would predict a preference for *was*, since it is only the plural ϕ−features that are acquired on the second cycle. An even stronger test would be to pit frequency or retrieval-based pressures against agreement, for instance by comparing the presence/absence of third-singular *-s* on non-copular verbs (*There seem(*s) to be some toys in the box*).

7.3 Implications for syntactic theory

Our results offer corroborating evidence, from a new empirical domain, for differentiating canonical and non-canonical agreement constructions in terms of syntactic complexity. These findings show that the relevant hidden elements, rules and operations are not just formalizations over ideal data, but are psychologically real, with consequences for learning, development and implementation.

It is nevertheless an open question how we should link complexity to errors. One possibility is that the link lies in the execution of more complex structures.[10] This would be a direct link if increased syntactic complexity has a progressive cost, and an indirect link if some other capacity of the mind that supports syntactic construction has that progressive cost. But it is also possible that complex structures are simply be harder to learn. This could either be because the learning a second agreement step has as its prerequisite the learning of the first, or alternatively, because the hypothesis space for multi-step agreement patterns is just larger.

A final connection worth highlighting is the one to typology. Across languages, we see a recurring pattern: full agreement with preverbal subjects, but no agreement with postverbal ones. We see this pattern in Fiorentino, Trentino, colloquial European Portuguese, Standard Arabic, among others. Consider Fiorentino in (17).

(17) Fiorentino (Brandi & Cordin (1989):122)

a. *glì* *ha* *telefonato* *delle* *ragazze*
3sg.m have.3sg phone some girls.3pl.f
'Some girls have telephoned'

b. **le* *hanno* *telefonato* *delle* *ragazze*
3pl.f have.3pl phone some girls.3pl.f

In this language, verbs typically agree with the subject in person, number, and gender — except when the subject appears to the right of the verb. In (17-a), the subject "some

[10]It should be noted though that similar "simple" linking hypotheses have had, at best, mixed success. For instance, the Derivational Theory of Complexity (Miller & Chomsky, 1963), which tried to map the number of syntactic transformations in a sentence onto processing difficulty, has not been well-supported by evidence.

girls" is feminine and plural, yet instead of feminine plural agreement, we get the default form, 3rd person singular masculine. Forcing feminine plural agreement, in fact, results in ungrammaticality (17-b).

This pattern looks remarkably similar to what we observed in child English: agreement is successful with a preverbal nominal, but not with a postverbal one. Fiorentino data are complicated and lend themselves to multiple analyses, but one account is particularly intriguing given the data from child English (Brandi & Cordin (1981), Rizzi (1986), Suñer (1992), Cardinaletti (2003); cf. Baier (2018)). This account posits a null expletive in the subject position of these inversion constructions. If so, there would be a striking syntactic parallel between child English and a set of unrelated subject-inversion languages.

This is a familiar story in developmental linguistics: even when child grammars diverge from target, their deviations tend to reflect typologically attested patterns (see e.g., Hyams (1986), Thornton (1990), Déprez & Pierce (1993), Deen (2002), Crain (2008), Snyder (2001), Yatsushiro & Sauerland (2018), and many others). This is reassuring from a perspective on child language as, first and foremost, a human language — albeit uniquely transient. On such a view, child language does not just tell us how knowledge of a first language grows, but also can help uncover generalizations about language as a whole.

8 Data Availability

All data and analysis code for the studies reported here can be found here: https://osf.io/k8nge/?view_only=85ddd414f1e241e9bc022a2f8bdae0dd.

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